3-Week Robotics Training Program (Elegoo Super Starter Kit) (1st - 19th December 2025)

Day	Topics / Activities	Learning Objectives	Tools / Assignments
Day 1	Intro to Robotics: Definition and Branches Expectations Basic computer skills PictoBlox basics	Understand robotics; Learn block coding; Build simple logic	PictoBlox activity; Student reflection
Day 2	Electronics basics; Breadboard; LEDs & resistors	Build simple circuits; Understand polarity; Control LEDs	LED patterns challenge; TinkerCAD/Fritzing intro
Day 3	Arduino IDE intro Digital & Analog output LED fade	Upload sketches Use PWM Control LED	LED fade TinkerCAD Simulation
Day 4	Inputs: Buttons, LDR, Potentiometer; Serial Monitor	Use digitalRead() & analogRead() Build input-driven systems	Touch/light-controlled system; Fritzing schematic
Day 5	Project ideation Mini prototype Soft skills (teamwork, pitching)	Brainstorm problems Work in teams Practice communication	Mini project 2-minute pitch
Day 6	Motors & Drivers: DC motor (Motor driver) , Servo, Stepper + ULN2003	Drive motors Control direction & speed	Servo gate DC motor control TinkerCAD motor simulation
Day 7	Sensors: Ultrasonic, DHT11, tilt sensor	Read sensor data Build automation systems	Motion alarm; Distance meter; Data logging
Day 8	Displays: LCD1602, 7-segment,	Display text & numbers on LCD and 7 segment displays	LCD sensor display Fritzing diagram
Day 9	IR Remote-Controlled Systems + Multi-Device Control Reading IR codes Mapping buttons to actions Combining IR remote with LEDs, buzzer, motors, LCD	Decode IR signals using Arduino Map IR commands to device functions Build a remote-controlled	IR-controlled system

		robotic/electronic system	
Day 10	Project ideation; System integration; Presentation	Combine modules; Design system flow; Improve teamwork	Draft project proposal; System flowchart
Day 11	Smart robotics; Code optimization; System simulation	Develop efficient code; Simulate systems	Smart device build; Full TinkerCAD simulation
Day 12	User interfaces Keypad + LCD menus Buzzer alerts	Build human-robot interface; Multi-input logic	Keypad+LCD menu system
Day 13	Project ideation; Documentation preparation	Develop presentation skills; Document circuits/code	Fritzing final diagram
Day 14	Final project build day	Apply all skills; Assemble final system	Working prototype; Testing checklist
Day 15	Final presentation & demo day	Demonstrate complete robotics workflow; Present final solution	Pitch deck; Robot demo; Final report